SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

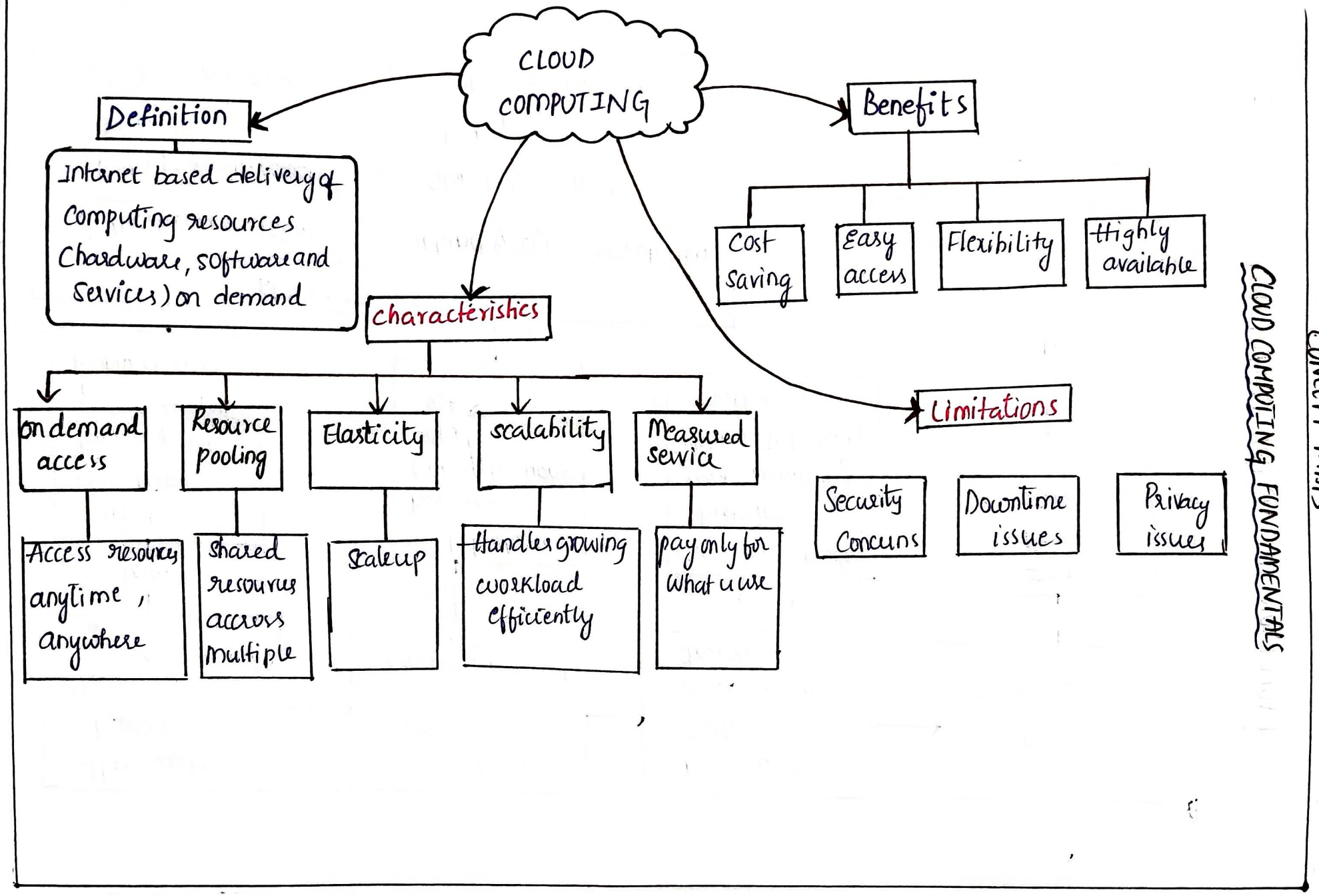
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

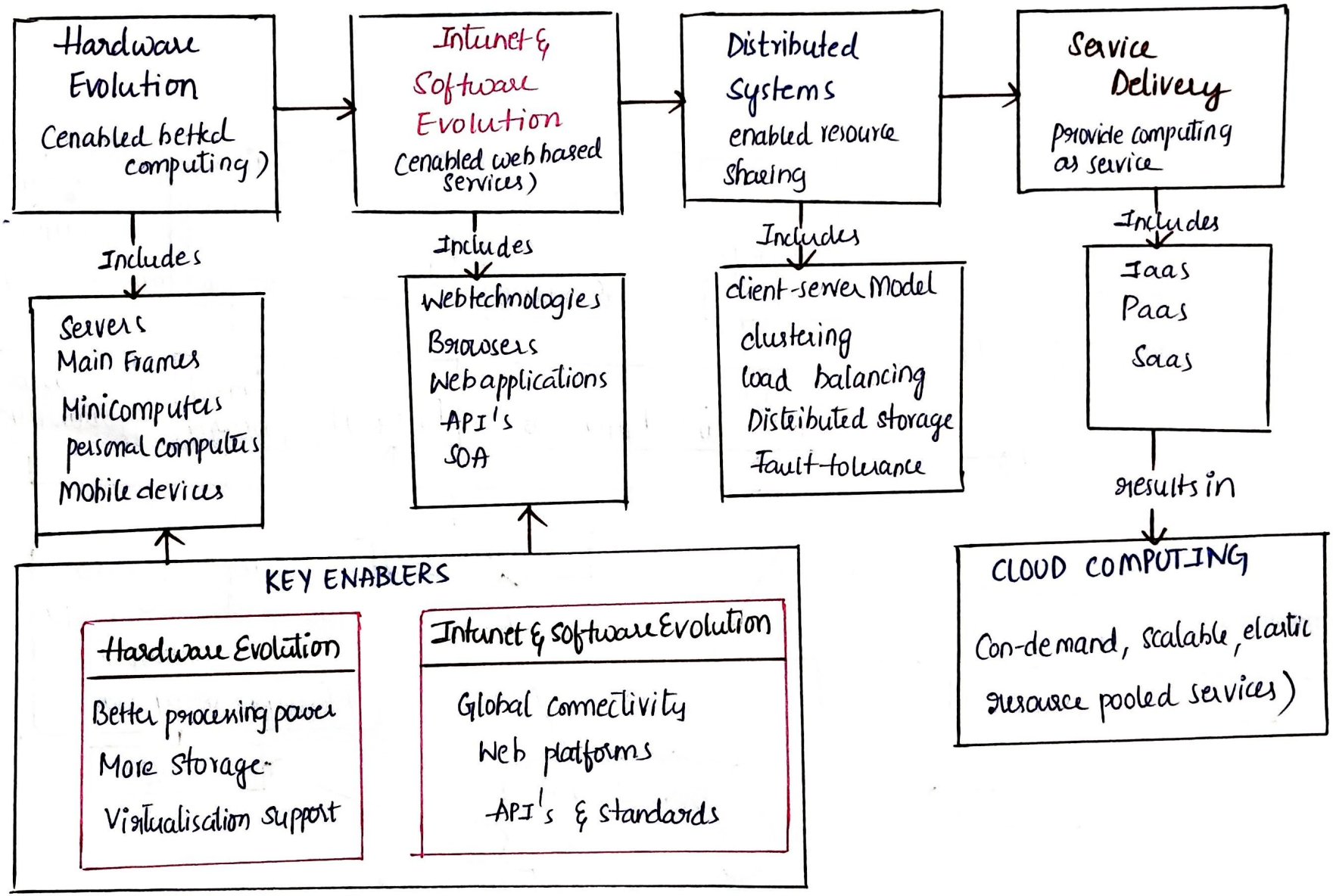
Concept Mapping Rubric

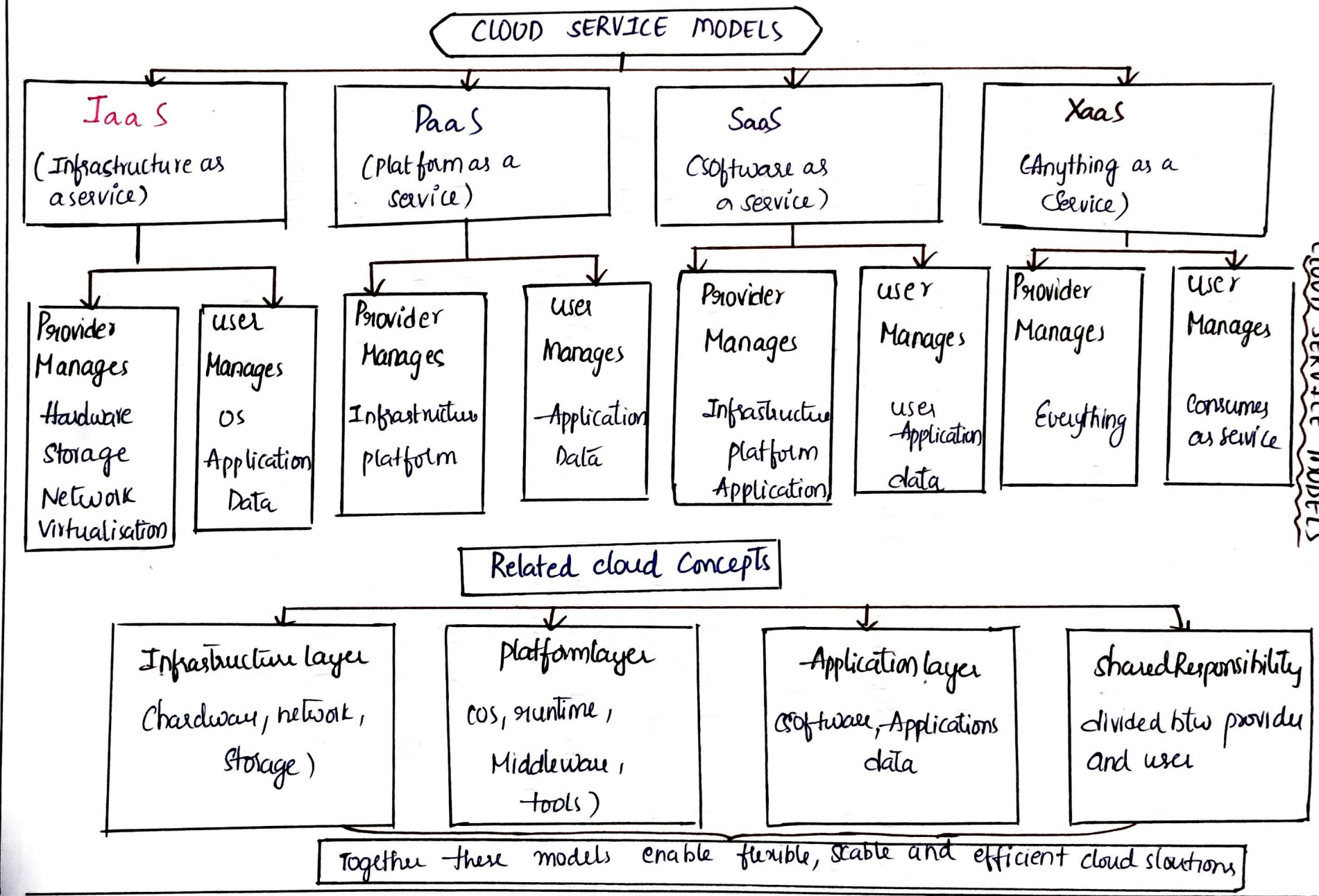
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented


CLOUD COMPUTING
Definition
Internet based delivery of computing resources (hardware, software and services) on demand
Benefits
Cost saving
Easy access
Flexibility
Highly available
characteristics
on demand access
Access resources anytime, anywhere
Resource pooling
Shared resources access multiple
Elasticity
Scale up
scalability
Handles growing workload efficiently
Measured servic
pay only for what u use
Limitations
Security Concuns
Downtime issues
Privacy issues

EVOLUTION OF CLOUD COMPUTING




CLOUD SERVICE MODELS
IaaS
(Infrastructure as a service)
PaaS
(Platform as a service)
SaaS
(Software as a service)
XaaS
(Anything as a Service)
Provider Manages
Hardware
Storage
Network
Virtualisation
User Manages
OS
Application
Data
Provider Manages
Infrastructure
platform
User Manages
Application
Data
Provider Manages
Infrastructure
platform
Application
User Manages
user Application
data
Provider Manages
Everything
User Manages
Consumes as service
CLOUD SERVICE MODELS
Related cloud Concepts
Infrastructure layer
(Hardware, network, Storage)
Platform layer
(OS, runtime, Middleware, tools)
Application layer
(Software, Applications data
Shared Responsibility
divided btw provider and user
Together these models enable flexible, scable and efficient cloud solutions

4. VIRTUALIZATION AND WEB SERVICES AS ENABLERS OF CLOUD

